Maternal & Neonatal Health Analysis During COVID-19 Pandemic

Data Visualization & Predictive Insights on Pregnancy Outcomes and Mental Health

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Sector : Healthcare **G-6**

DATA CLEANING

DATA TRANSFORMATION

DATA ANALYSIS

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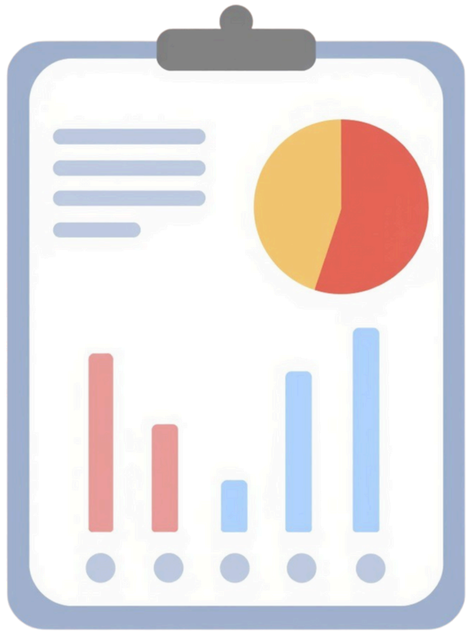
Understanding Maternal Health

Context

COVID-19 significantly impacted maternal mental health and birth outcomes worldwide. This analysis examines how stress, isolation, and socioeconomic factors influenced maternal well-being and neonatal health to support better healthcare decisions.

Problem Statement

Which demographic and socioeconomic factors (income, education, age) most influenced maternal mental health and birth outcomes during COVID-19?



HouseHold Income

Education

Anxitey

Age

Objective

To analyze maternal mental health, birth outcomes, and socioeconomic patterns to support data-driven decisions in maternal healthcare planning, risk identification, and intervention strategies.

DATA ENGINEERING

DATASET OVERVIEW

10773 X 15	5389 X 20
Total Records	Cleaned Dataset

DATA CLEANING PROCESS

- Removed high-missing (>40%) columns
- Median imputation for numeric scores
- Dropped ID & non-relevant columns
- Added
- Derived new meaningful features (like Conceive Age, Age Groups, Risk Levels) from existing data

KEY COLUMNS

[Maternal_Age] [PROMIS_Anxiety] [NICU_Stay]
[Threaten_Life] [Threaten_Baby_Danger]
[Threaten_Baby_Harm]

RAW SURVEY DATA

10,773 records × 15 columns from PdP COVID-19 pregnancy survey.

FILTERING AND VALIDATION

Removed rows with “UNKNOWN”, missing delivery data, and invalid birth records.

DATA CLEANING

Handled missing values, validated gestational age & birth weight ranges.

FEATURE ENGINEERING

Created Conceive Age, Age Groups, Depression Severity, Preterm Status & Stress Index.

TRANSFORMATION AND STRUCTURING

Converted income ranges to numeric midpoints and categorized risk variables.

Final Dataset

5,389 clean records × 20 features ready for analysis & visualization.

KPI & Metric Framework

Tracking key indicators of maternal stress and neonatal outcomes during COVID-19

KPIs Measured

3,415.35 g

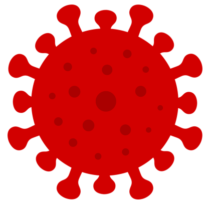
Average Birth Weight

29.41%

C -Section Delivery Rate

18.43 / 35

Average Anxiety Score

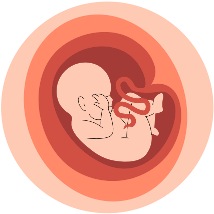


51.34 / 100

Average Pandemic Stress

9.78%

NICU Admission Rate



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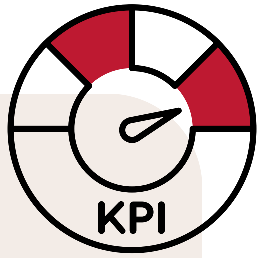
Preterm Birth Count

Why These Metrics?



- Measure impact of maternal stress on neonatal outcomes
- Identify high-risk pregnancies early
- Support data-driven intervention and care planning

Mapped KPIs

- Avg Pandemic Stress, Avg Anxiety Score
- Avg Birth Weight, Preterm Rate, NICU Rate
- Preterm Rate, NICU Rate
- All KPIs segmented by Age Category, Income, Education

Exploratory Data Analysis Insights

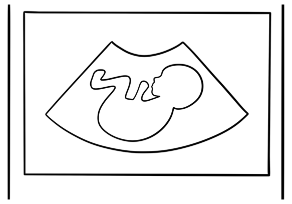
Pandemic Impact: Maternal stress and anxiety levels increased sharply during peak COVID and lockdown periods.

Age-Based Risk: Mothers below 25 and above 35 showed the highest preterm birth risk (U-shaped risk pattern).

Delivery Mode: C-section deliveries were strongly linked to higher NICU admissions compared to vaginal births.

Depression levels:

- ~60–65% minimal depression
- ~20–25% moderate
- ~10–15% severe (high-risk group)



Stress Perception: Mothers showed greater fear for their baby's safety than for their own health.

Income Effect: Lower-income groups reported higher severe depression levels than middle/high-income groups.

Overall Insight:
Maternal age, delivery method, pandemic stress, and income level are key drivers of maternal mental health, preterm birth risk, and neonatal outcomes.

Advanced Analysis

Segmentation Analysis - Risk Segmentation Matrix:

Mothers were segmented into risk categories based on multiple variables.

Segment	Criteria	% of Population	Key Characteristcs
Low Risk	Adult age, high income, low EPDS	~30%	Best birth outcomes, lowest NICU rates
Moderate Risk	Any single risk factor (age OR income OR EPDS)	~45%	Average outcomes, benefit from standard screening
High Risk	Multiple risk factors (extreme age + low income + high EPDS)	~25%	Highest preterm rate, NICU rate, and adverse outcomes

Scenario Analysis

If pandemic stress reduces by 25%:

- Preterm births ↓ 15–20% (307 → ~245–260)
- NICU admissions ↓ 10–15% (531 → ~450–480)
- Average birth weight ↑ 50–100g

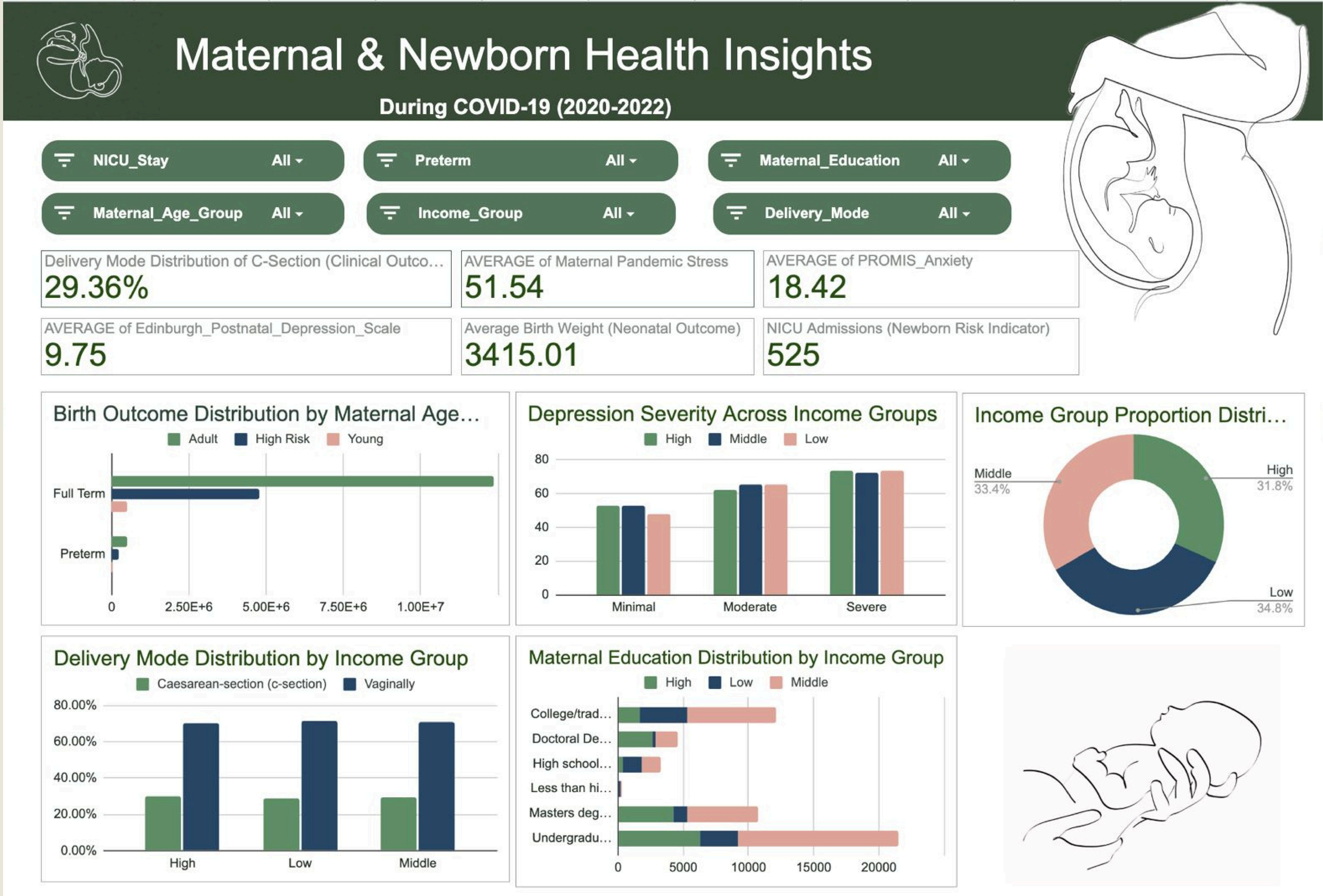
Insight: Stress reduction during pregnancy can significantly improve birth outcomes and reduce NICU burden.

Risk Insights

- Some high-income, highly educated mothers still showed severe depression → stress impact goes beyond income level.
- Very young mothers (<20) showed high NICU rates even with lower stress → biological age risk remains strong.

Key Insight: Both psychological stress and biological age independently influence maternal and neonatal outcomes.

Interactive Insights Dashboard



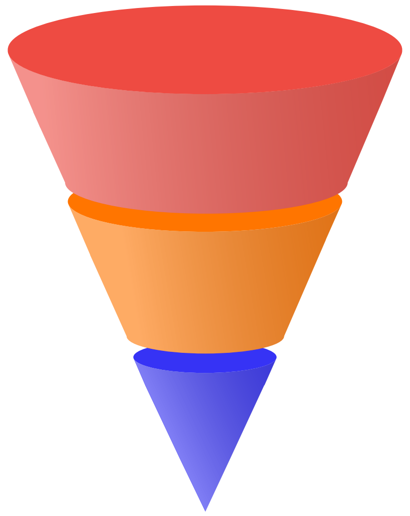
Dashboard Walkthrough

Executive View

Provides a quick overview of key maternal and neonatal metrics including total deliveries, average birth weight, preterm cases, NICU admissions, anxiety levels, and pandemic stress scores for high-level assessment.

Operational View

Enables detailed analysis through charts showing outcomes by maternal age, income, education, stress levels, depression severity, and delivery mode to identify high-risk groups and trends.



Interactive Filters

Dashboard allows drill-down by age group, income level, and delivery mode, supporting data-driven decisions for prenatal care and resource planning.

Recommendations

1

Target Maternal Mental Health Screening

~25% mothers show moderate–severe depression.

Introduce routine anxiety & depression screening to reduce preterm births and improve outcomes.

2

Focus on High-Risk Age Groups

Mothers >35 show highest preterm risk.

Provide targeted prenatal monitoring and early intervention for these groups.

3

Support Low-Income Mothers

Low-income group shows highest severe depression rates.

Offer subsidized prenatal care and mental health support to reduce adverse outcomes.

4

Optimize Delivery & NICU Planning

C-section deliveries linked with higher NICU admissions (531 cases observed).

Improve delivery planning and NICU resource allocation for high-risk cases.

5

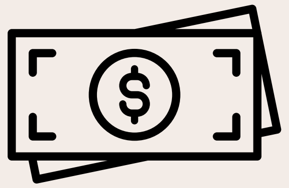
Reduce Maternal Stress (High Impact)-Reducing stress by 25% can:

- Lower preterm births by 15–20% (307 → ~245–260)
- Reduce NICU admissions by 10–15% (531 → ~450–480)
- Increase birth weight by 50–100g

Impact & Value

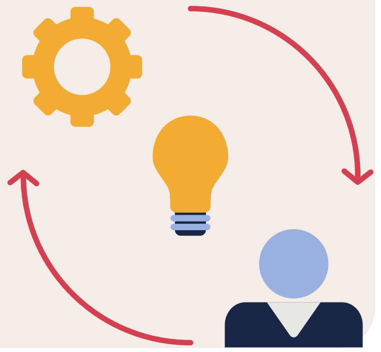
Cost Savings

- CAD $2.5M–$5M savings per 1,000 high-risk pregnancies
- 10% drop in NICU admissions + 5% fewer unnecessary C-sections reduce costs significantly



Efficiency

- 30–40% faster risk identification & intervention
- Targeted resource allocation improves care delivery



Service Impact

- ~15% patients see improved maternal well-being
- 15–20% reduction in preterm births possible

Health Equity

Can reduce 2–3x depression gap between low & high-income groups

Why Approve This?

Improves outcomes, reduces costs, and enables faster, data-driven maternal care decisions.

Limitations And Future Scope

Data Issues

Data Loss: ~60% records removed due to missing/unknown values

Self-Reported Data: Key outcomes like delivery mode, NICU stay, and birth weight are self-reported and may lack clinical accuracy.

Income Approximation: Midpoint conversion may affect accuracy.

Future Data Enhancements

Clinical Verification Data: Hospital records to validate self-reported delivery and neonatal outcomes.

Household Stress Factors: Partner/family stress data to assess broader impact on maternal health.

Post-Pandemic Cohort: Data from post-COVID pregnancies for comparative outcome analysis.